BAM-U-Net: A Bidirectional Adversarial Mamba U-Net for Large-Scale EEG Foundation Modeling

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Abstract

Foundation models have recently transformed EEG decoding. However, most existing approaches rely on task-specific fine-tuning to adapt latent representations and often underperform in linear probing settings, indicating limited feature generalization. Moreover, the quadratic computational complexity of standard attention mechanisms with respect to sequence length makes these models impractical for long-term EEG monitoring. To address these limitations, we introduce BAM-U-Net (Bidirectional Adversarial Mamba U-Net), a scalable EEG foundation model built upon the Mamba-2 architecture. BAM-U-Net integrates BiMamba-2 blocks within a U-Net backbone, enabling efficient multi-scale temporal feature extraction while preserving long-range dependencies in neural signals. Critically, it introduces an adversarial denoising objective that enhances reconstruction quality and enables the model to capture fine-grained, high-frequency, and physiologically realistic neural patterns. Extensive experiments demonstrate that BAM-U-Net outperforms seven state-of-the-art foundation models under linear probing on five out of six tasks, while matching performance on the remaining task. These results highlight strong feature transferability without task-specific fine-tuning.

1 Introduction

Electroencephalography [1] (EEG) is a widely used non-invasive, low-cost modality for recording brain activity and building Brain-Computer Interfaces [2, 3] (BCIs). Analyzing these signals enables neurologists to interpret neuronal activity and underlying brain processes.

Recent Foundation Models [4, 5, 6, 7, 8, 9] leverage Transformers [10] to extract universal EEG features. However, due to EEG's low signal-to-noise ratio and non-stationarity [11], these models typically require resource-intensive fine-tuning to adapt to new tasks. While linear probing [12]—training only a linear head on a frozen encoder—offers an efficient alternative, it currently yields suboptimal results on existing models.

Many EEG Foundation Models rely on Transformer-based models which introduce quadratic complexity with respect to the sequence length. This limits their scalability for long-term clinical monitoring, such as sleep staging or seizure detection, where recordings can span several hours [13, 14]. To mitigate this, State-Space Models [15] (SSMs), and more recently Mamba-based architectures [16, 17], including EEG-specific variants such as LuMamba [18], have emerged as a promising linear-time alternative for long-sequence modeling. However, their potential for high-fidelity EEG reconstruction and universal feature extraction remains largely under-explored compared to their Transformer counterparts.

Anonymous code: https://anonymous.4open.science/r/BAM-U-Net/

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

Most Foundation Models rely predominantly on the Masked Auto-Encoder [19] (MAE) paradigm, where the primary objective is to reconstruct masked portions of the signal. EEGPT [5] combines a standard reconstructive loss with an auxiliary contrastive loss, while REVE [4] and LUNA [9] employ a standard reconstructive pixel-wise loss as their primary objective. While the reconstruction objective forces the model to learn useful latent features, recent Foundation Models seldom refine this objective and prefer to add auxiliary tasks to shape the latent space. Balancing multiple losses can be tricky to stabilize within a single model if they are not perfectly aligned with one another.

To address these challenges, we introduce BAM-U-Net (Bidirectional Adversarial Mamba U-Net), an EEG Foundation Model based on an asymmetric U-Net [20] architecture designed for the extraction of high-fidelity and robust EEG features. The contributions are threefold:

- **Adversarial Denoising Objective.** Unlike traditional Masked Auto-Encoders that rely solely on pointwise reconstruction, we integrate a patch-based adversarial discriminator. This objective forces the encoder to capture physiologically plausible high-frequency signatures that are often smoothed out by standard pixel-wise losses, significantly enhancing the expressive power of the learned representations.
- **State-of-the-art Linear Probing Performance.** We show that our unified adversarial reconstruction framework produces highly discriminative off-the-shelf representations. Evaluated across six diverse downstream tasks, BAM-U-Net consistently outperforms current foundation models in the linear-probing regime, ranking first on five tasks and second on the remaining one, effectively bridging the gap between frozen-encoder efficiency and task-specific accuracy.
- **Linear-time Sequence Modeling.** By leveraging a BiMamba-2 backbone, our model achieves linear complexity with respect to sequence length. This allows BAM-U-Net to process high-resolution raw EEG recordings without the memory constraints or the need for temporal patching inherent to Transformer-based architectures.

2 Method

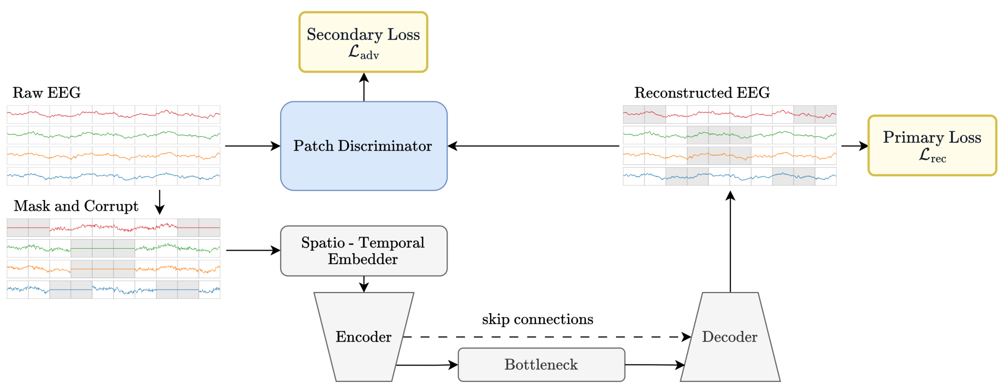

Figure 1: **Overview of the BAM-U-Net pre-training framework.** The raw continuous EEG signal is corrupted with noise and partially masked before being projected by a multi-branch temporal embedder. An asymmetric BiMamba-2 U-Net processes the sequence to extract a latent representation (bottleneck) and decodes it to reconstruct the full waveform. The model is optimized jointly via a primary denoising reconstruction loss ($\mathcal{L}_{\text{rec}}$) and a secondary adversarial loss ($\mathcal{L}_{\text{adv}}$), enabling the extraction of multi-scale, robust temporal features. GAP denotes global average pooling.

2.1 Background

We build on Mamba-2, a selective state space model (SSM) designed for linear-time sequence modeling [16, 17]. For an input sequence $X = (x_1, \dots, x_T)$ with $x_t \in \mathbb{R}^D$, a selective SSM updates a latent state $h_t \in \mathbb{R}^N$ as

$$h_t = \bar{\mathbf{A}}_t h_{t-1} + \bar{\mathbf{B}}_t x_t, \qquad y_t = \mathbf{C}_t h_t, \tag{1}$$

where the input-dependent parameters $\bar{\mathbf{B}}_t$, $\mathbf{C}_t$, and the discretization used to form $\bar{\mathbf{A}}_t$ allow the model to adapt its dynamics to the current token. Unlike self-attention, the recurrence in Eq. (1) is scanned with linear complexity in the sequence length, which is well suited to long EEG windows.

Because our setting uses fixed-length offline windows rather than causal streaming, we use a bidirectional variant. Let $\mathcal{M}(\cdot)$ denote one pre-normalized residual Mamba block,

$$\mathcal{M}(X) = \mathrm{Mamba2}(\mathrm{LN}(X)) + X. \tag{2}$$

We then adopt BiMamba-2 with sum fusion [21]:

$$\mathrm{BiMamba}(X) = \mathcal{M}_{\mathrm{fwd}}(X) + \mathrm{rev}(\mathcal{M}_{\mathrm{bwd}}(\mathrm{rev}(X)))\,, \tag{3}$$

where $\mathrm{rev}(\cdot)$ reverses the sequence along time. This gives each representation access to both past and future temporal context while preserving linear-time inference.

2.2 EEG representation and denoising objective

A pre-training sample is a clean EEG window $x \in \mathbb{R}^{C\times T}$, with $C = 19$ channels and T time samples. Unlike patch-tokenized EEG models [5, 6, 7, 8], our encoder operates on the full-resolution continuous signal. We introduce temporal patches only to define structured corruptions. Let $P = 200$ denote the patch length, corresponding to one-second windows, and let $N = T/P$ be the number of temporal patches. We therefore view the same window as $x^{\mathrm{patch}} \in \mathbb{R}^{C\times N\times P}$.

During pre-training, we first corrupt the input with additive Gaussian noise:

$$\tilde{x} = x + \sigma\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \mathbf{I}), \qquad \sigma = 0.2. \tag{4}$$

The decoder is trained to reconstruct the clean signal x, not the noisy input $\tilde{x}$. This turns masked reconstruction into a denoising objective [22] and improves robustness to acquisition-specific noise.

2.3 Multi-scale spatio-temporal masking and embedding

We build a binary mask $M \in \{0,1\}^{C\times N}$ on the channel-by-time-patch grid, where $M_{c,n} = 1$ means that patch (c, n) is hidden. Given a target masking ratio r, we iteratively sample channel–time rectangles until at least rCN entries are masked. The rectangle height h is sampled uniformly from $\{1, \ldots, C-1\}$, while the temporal width is sampled from a height-dependent range:

$$w \sim \mathcal{U}\{1, \ldots, w_{\max}(h)\}, \qquad w_{\max}(h) = \left\lfloor \frac{rCN}{3h} \right\rfloor \tag{5}$$

Equation (5) makes tall spatial masks narrow in time, while shorter masks can cover longer temporal spans, with the factor 3 preventing a single block from dominating the mask and encouraging at least three sampled blocks. This produces corruptions that are harder than independent point masking but still preserve enough local structure for stable reconstruction. The realized masking ratio may slightly exceed r because the final sampled block is kept rather than truncated.

The patch-level mask is then expanded back to sample resolution, yielding $\bar{M} \in \{0,1\}^{C\times T}$, and the actual encoder input becomes

$$x^{\mathrm{in}} = (1 - \bar{M}) \odot \tilde{x}. \tag{6}$$

The mask is therefore defined on patches, but the encoder consumes the continuous masked waveform $x^{\mathrm{in}} \in \mathbb{R}^{C\times T}$.

Before entering the encoder, the corrupted signal is mapped to $e \in \mathbb{R}^{32\times T}$ by a multi-branch temporal embedder. The embedder applies seven temporal convolutions with kernel sizes $\{1, 3, 7, 11, 51, 101, 201\}$, concatenates their outputs, and fuses them with a 1×1 convolution. Since each convolution mixes all input electrodes, every latent channel is a learned spatio-temporal combination of the original 19-channel montage.

2.4 Asymmetric encoder-decoder

The embedded sequence is processed by a U-Net-like BiMamba encoder-decoder [20]. The architecture is intentionally asymmetric: most of the capacity is placed in the encoder because only the embedder and encoder are retained for downstream tasks. This yields 13.3M encoder parameters versus 6.8M decoder-side parameters, including the bottleneck.

Let $z^{(0)} = e$. The encoder has four stages with widths

$$d_s = 2^{s-1}B, \qquad s \in \{1, 2, 3, 4\}, \tag{7}$$

so that $(d_1, d_2, d_3, d_4) = (32, 64, 128, 256)$. Each stage applies a projection layer followed by a stack of three BiMamba blocks:

$$u^{(s)} = \mathcal{B}_s\Big(p_s(z^{(s-1)})\Big), \tag{8}$$

where p_1 is a linear layer applied independently at each time step and p_s for $s > 1$ is a 1D convolution with kernel size 3. For $s < 4$, we then downsample time with max-pooling:

$$z^{(s)} = \begin{cases} \mathrm{Pool}(u^{(s)}) & \text{if } s < 4, \\ u^{(s)} & \text{if } s = 4. \end{cases} \tag{9}$$

After the encoder, the input is mapped from $\mathbb{R}^{32 \times T}$ to $z^{(4)} \in \mathbb{R}^{256 \times T/8}$.

The bottleneck applies a lightweight linear–BiMamba–linear transformation at the coarsest temporal resolution. The decoder then reconstructs the signal through three upsampling stages. At stage s, the current representation is linearly interpolated to the resolution of the corresponding encoder feature map, concatenated with the skip connection $u^{(s)}$, and processed by a 1D convolution followed by three BiMamba blocks. A final 1×1 convolution maps the last decoder state back to $\hat{x} \in \mathbb{R}^{C \times T}$.

Training minimizes masked denoising reconstruction over both hidden and visible samples:

$$\mathcal{L}_{\text{hid}} = \frac{\big\|\bar{M} \odot (\hat{x} - x)\big\|_2^2}{\|\bar{M}\|_1}, \qquad \mathcal{L}_{\text{vis}} = \frac{\big\|(1 - \bar{M}) \odot (\hat{x} - x)\big\|_2^2}{\|1 - \bar{M}\|_1}. \tag{10}$$

We then define the reconstruction objective as

$$\mathcal{L}_{\text{rec}} = \lambda_{\text{hid}} \mathcal{L}_{\text{hid}} + \lambda_{\text{vis}} \mathcal{L}_{\text{vis}}, \tag{11}$$

with $\lambda_{\text{hid}} = \lambda_{\text{vis}} = 1$ by default. In practice, this decomposition was more stable in our setting than more elaborate spectral reconstruction losses.

2.5 Adversarial network

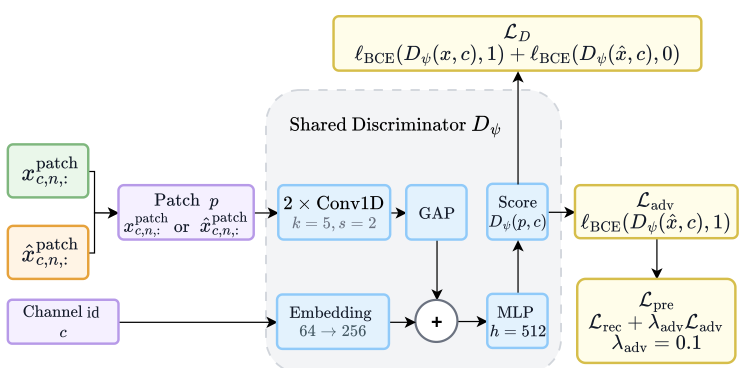

Figure 2: **Detailed schematic of the patch-based adversarial network.** The shared discriminator D_ψ processes real (clean) and fake (reconstructed) patches exclusively from masked locations, incorporating electrode-specific channel embeddings to account for spatial variability. By forcing the generator to fool the discriminator, this adversarial objective ensures the model captures physiologically plausible, high-frequency EEG signatures that are typically smoothed out by standard pointwise reconstruction. GAP denotes global average pooling.

To further encourage locally plausible reconstructions, we augment the denoising objective with the patch discriminator shown in Figure 2, trained in an adversarial fashion. The discriminator operates on the same temporal patching used by the masking module. Let $P = 200$ denote the patch length and let

$$x^{\text{patch}}, \hat{x}^{\text{patch}} \in \mathbb{R}^{C\times N\times P} \tag{12}$$

be the clean and reconstructed signals split into channel-wise patches.

The discriminator receives two matched sets of examples defined on the masked locations. Real examples are the corresponding clean patches from locations selected by the mask,

$$\mathcal{R} = \left\{ \left(x^{\text{patch}}_{c,n,:}, c \right) : M_{c,n} = 1 \right\}, \tag{13}$$

while fake examples are the reconstructed patches at those same locations,

$$\mathcal{F} = \left\{ \left(\hat{x}^{\text{patch}}_{c,n,:}, c \right) : M_{c,n} = 1 \right\}. \tag{14}$$

Using the same masked index set for both classes keeps the class sizes identical and concentrates the adversarial objective on patches whose content must be predicted rather than directly observed.

For each single-channel patch $p \in \mathbb{R}^P$ and channel index c, the discriminator applies two temporal convolutions with kernel size 5 and stride 2, expanding the feature width from 1 to 32 and then to 256. Each convolution is followed by LeakyReLU and dropout, and global average pooling produces a 256-dimensional patch descriptor. To account for electrode-specific waveform statistics, the discriminator also learns a 64-dimensional channel embedding, projects it to 256 dimensions, and adds it to the patch descriptor before a two-layer MLP with hidden width 512 outputs the real/fake logit $D_\psi(p, c)$.

We use the standard binary cross-entropy GAN objective with logits [23]. Denoting by $\ell_{\text{BCE}}(s, y)$ the binary cross-entropy between logit s and label $y \in \{0, 1\}$, the discriminator objective is

$$\mathcal{L}_D = \frac{1}{2}\mathbb{E}_{(p,c)\sim\mathcal{R}} \left[\ell_{\text{BCE}}(D_\psi(p, c), 1)\right] + \frac{1}{2}\mathbb{E}_{(p,c)\sim\mathcal{F}} \left[\ell_{\text{BCE}}(D_\psi(p, c), 0)\right]. \tag{15}$$

When updating the reconstruction network, the discriminator is frozen and the generator is trained to fool it on fake patches:

$$\mathcal{L}_{\text{adv}} = \mathbb{E}_{(p,c)\sim\mathcal{F}} \left[\ell_{\text{BCE}}(D_\psi(p, c), 1)\right]. \tag{16}$$

The full pre-training objective is therefore

$$\mathcal{L}_{\text{pre}} = \mathcal{L}_{\text{rec}} + \lambda_{\text{adv}}\mathcal{L}_{\text{adv}}, \qquad \lambda_{\text{adv}} = 0.1. \tag{17}$$

The reconstruction network is optimized with $\mathcal{L}_{\text{pre}}$, while the discriminator is optimized with $\mathcal{L}_D$. This auxiliary adversarial signal encourages the decoder to produce locally realistic EEG waveforms rather than merely minimizing pointwise reconstruction error.

3 Experiments

3.1 Pre-training dataset preprocessing

Pre-training Data. We leverage the Temple University Hospital EEG (TUEG) Corpus [24], the largest publicly available repository of clinical EEG, comprising 27,062 hours of recordings. To ensure the integrity of our pre-training phase and avoid data leakage, we strictly exclude any recordings belonging to the TUAB or TUEV subsets, which are reserved for downstream evaluation.

Preprocessing Protocol. To maintain consistency with recent large-scale EEG models and ensure a high-quality signal for Mamba-2 training, we implement the robust preprocessing pipeline established by STELAR [25]. Following this protocol, we first discard the initial and final minutes of each session to eliminate non-stationary artifacts common at recording boundaries. We restrict our spatial coverage to the 19 most frequent electrodes of the international 10-20 system [26], providing a standardized montage across heterogeneous acquisition setups.

The signals undergo a band-pass filter (0.3–75 Hz) and a notch filter at 60 Hz to suppress power-line interference. All data are resampled to 200 Hz and segmented into non-overlapping 30-second windows. Following [25, 7], we apply a strict artifact rejection threshold, removing any samples with a peak-to-peak amplitude exceeding 100 μV. Finally, the remaining windows are Z-score normalized.

3.2 Downstream tasks

Downstream setup and metrics Once training is completed, we discard the masking mechanism, the bottleneck, the decoder, and the discriminator. The model used for downstream tasks therefore consists only of the input embedding layer and the encoder, and transforms the input $x \in \mathbb{R}^{C,T}$ into features $f \in \mathbb{R}^{8 \cdot B, T/8}$.

Mirroring the downstream framework of EEGPT [5], we evaluate the model by performing classification on diverse tasks spanning various datasets. Since downstream datasets may have a different number and organization of channels compared to the pre-training data, we first apply a 1×1 convolution before the input embedding to map the downstream input $x_d \in \mathbb{R}^{C_d, T_d}$ to the expected input format $x \in \mathbb{R}^{C, T_d}$. After processing through the encoder, the resulting features are mapped to output logits using the same linear layer as in EEGPT [5]. We focus our downstream evaluation on the linear-probing regime: the pre-trained encoder is frozen, while only the channel adapter and the classification head are trained. This setting isolates the quality of the learned representations.

Downstream datasets and preprocessing To ensure a fair comparison with other baselines, we keep the same train, validation, and test partitions in our downstream tasks. Preprocessing is applied to the downstream data as specified in EEGPT [5], which ensures fair and reproducible comparison with standard baselines. Additional details are given in Table 12

Table 1: Overview of EEG datasets and their corresponding BCI tasks.

BCI Task	Dataset	Sampling Rate	#Channels	Duration	Labels
Motor Imagery	BCIC-2A	250 Hz	22	4s	4-class
	PhysioNet-MI	160 Hz	64	4s	4-class
Sleep Staging	Sleep-EDFx	100 Hz	2	30s	5-class
Event-related Potentials	KaggleERN	200 Hz	56	2s	2-class
Seizure / Event Detection	TUEV	250 Hz	23	5s	6-class
Abnormal EEG Detection	TUAB	250 Hz	23	10s	2-class

4 Results

4.1 Linear probing across downstream tasks

We first evaluate whether the frozen BAM-U-Net encoder provides linearly accessible information across heterogeneous EEG tasks. Balanced accuracy is the primary metric because several datasets are class-imbalanced, especially TUAB and TUEV. All models are evaluated with the same splits and the same linear-probing protocol summarized in Table 1; task-specific fine-tuning is therefore deliberately excluded from this comparison.

Table 2: Linear-probing balanced accuracy (%) across six downstream datasets. Best results are in bold and second-best results are underlined. Other metrics are reported in the appendix.

Model	BCIC-2A	PhysioNet-MI	KaggleERN	Sleep-EDFx	TUAB	TUEV	Avg.
CBraMod	27.53 ± 1.96	39.78 ± 5.62	50.02 ± 0.14	50.91 ± 11.62	64.55 ± 7.65	44.22 ± 1.22	46.17
CSBrain	47.19 ± 0.05	51.14 ± 3.25	54.28 ± 0.14	58.17 ± 1.43	76.01 ± 0.23	$\underline{52.36 \pm 1.00}$	56.53
EEGPT	$\underline{53.27 \pm 1.51}$	$\underline{54.67 \pm 2.21}$	$\underline{55.75 \pm 0.74}$	60.10 ± 1.72	$\mathbf{79.81 \pm 1.90}$	42.13 ± 1.22	$\underline{57.62}$
LaBraM	31.30 ± 0.04	27.82 ± 0.55	49.98 ± 0.01	62.52 ± 4.70	77.48 ± 0.07	44.09 ± 1.07	48.87
LUNA	33.73 ± 3.90	43.14 ± 2.16	51.08 ± 0.19	60.02 ± 4.20	69.89 ± 0.17	27.19 ± 1.10	47.51
LuMamba	31.69 ± 3.78	44.11 ± 2.17	54.26 ± 0.89	$\underline{65.65 \pm 3.45}$	73.56 ± 0.53	32.22 ± 3.32	50.25
REVE	51.81 ± 9.39	47.19 ± 4.50	55.20 ± 0.41	60.87 ± 4.72	72.74 ± 1.75	44.14 ± 3.23	55.33
BAM-U-Net	$\mathbf{60.58 \pm 2.80}$	$\mathbf{61.83 \pm 3.17}$	$\mathbf{56.41 \pm 0.34}$	$\mathbf{67.74 \pm 2.86}$	$\underline{79.28 \pm 0.81}$	$\mathbf{52.58 \pm 1.93}$	$\mathbf{63.07}$

Table 2 shows that BAM-U-Net ranks first on five of the six benchmarks and second on the remaining one. The gains are largest on motor imagery, where BAM-U-Net improves over the strongest baseline by 7.31 points on BCIC-2A and 7.16 points on PhysioNet-MI. These tasks require subject-robust discrimination of oscillatory sensorimotor patterns, suggesting that the adversarial denoising objective does not merely improve waveform reconstruction, but also yields features that a linear classifier can separate after transfer. BAM-U-Net also leads on Sleep-EDFx by 2.09 points over LuMamba and on KaggleERN by 0.66 points over EEGPT, while narrowly surpassing CSBrain on TUEV by 0.22 points. TUAB is the only exception: EEGPT remains the strongest single model by 0.53 points, although BAM-U-Net achieves the second-best score with a substantially smaller standard deviation. Averaged over the six balanced-accuracy columns, BAM-U-Net reaches 63.07%, compared with 57.62% for the strongest baseline average, EEGPT.

4.2 Geometry of frozen representations

To complement the aggregate metrics, Figure 3 visualizes the test-set geometry of frozen features on BCIC-2A after mean pooling. These projections are qualitative diagnostics rather than scoring metrics, since any two-dimensional embedding can distort high-dimensional distances. Nevertheless, the visual pattern is consistent with the quantitative results: several baselines spread all four motor-imagery classes through a shared region, whereas BAM-U-Net forms broader class-aligned structures, most visibly for classes 1 and 3. The remaining overlap is expected for BCIC-2A, but the separation is sufficient for a linear head to exploit, matching the margin observed in Table 2.

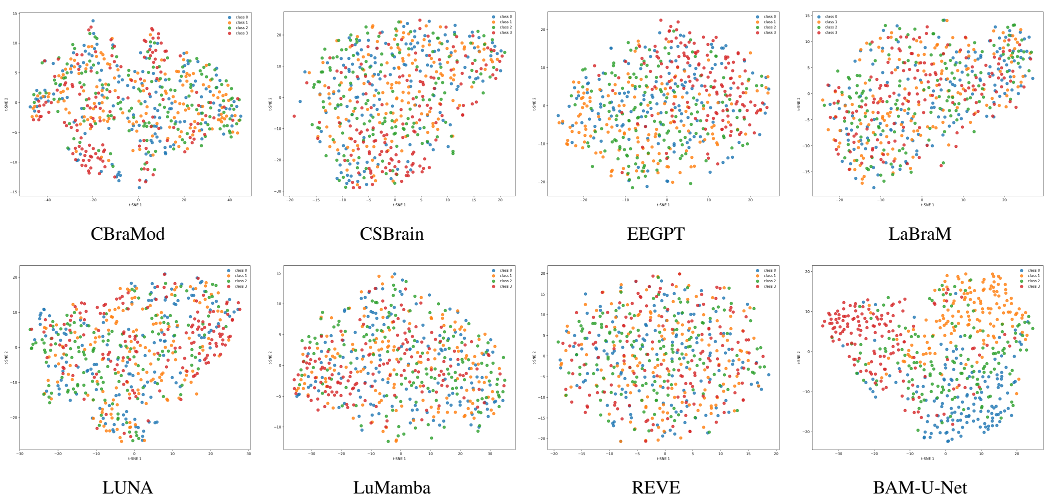

Figure 3: Low-dimensional projections of frozen encoder features on the BCIC-2A test set. Each point corresponds to a trial after mean pooling, and colors denote the four motor-imagery classes. BAM-U-Net produces more class-aligned geometry than the compared frozen encoders.

We also examine representation collapse on TUAB by measuring pairwise cosine similarity between flattened test embeddings. A distribution concentrated near one indicates that many examples share nearly the same direction in feature space, limiting the degrees of freedom available to a linear classifier. Figure 4 shows that BAM-U-Net avoids the highly collapsed regimes observed for several baselines. It has the lowest mean cosine similarity (0.517) and upper-tail similarity ($p_{95} = 0.653$; full statistics in Appendix Table 5), while CBraMod, LaBraM, and REVE all exceed 0.90 average similarity.

4.3 Effect of adversarial denoising

The main architectural difference between BAM-U-Net and a standard denoising autoencoder is the patch-level adversarial signal. Figure 5 isolates its effect by comparing reconstructions with and without the discriminator. The adversarially trained model reduces the displayed total reconstruction loss from 2.98 to 1.72 and more faithfully follows both the phase and amplitude of the raw signal outside the masked region as well as within the shaded reconstruction interval. Qualitatively, the non-adversarial model tends to smooth or drift through high-amplitude transients, whereas the adversarial model preserves sharper local waveform structure.

This improvement is also visible in the frequency domain. Figure 6 compares the power spectral density (PSD) of one reconstructed sample with and without the adversary. The adversarial reconstruction not only matches the signal more closely in time, but also places the dominant spectral

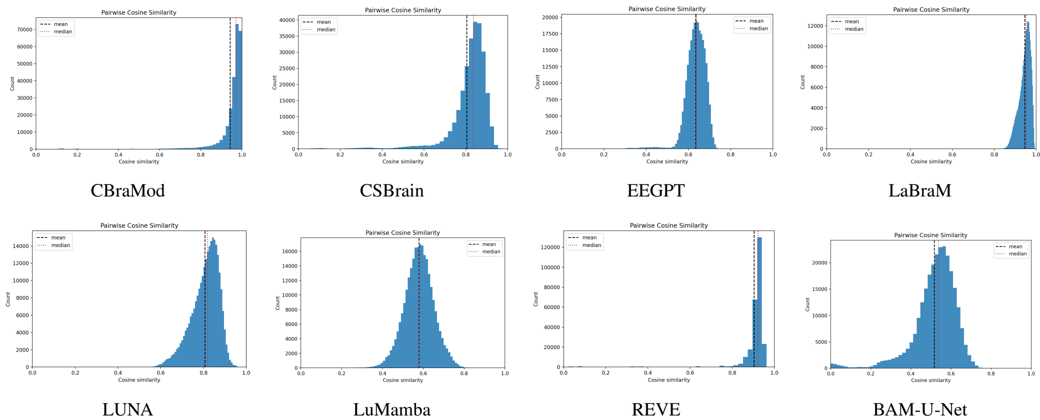

Figure 4: Pairwise cosine-similarity distributions of TUAB embeddings. Dashed black and dotted red lines denote the mean and median, respectively. Distributions shifted away from one indicate less directional collapse of the frozen representation.

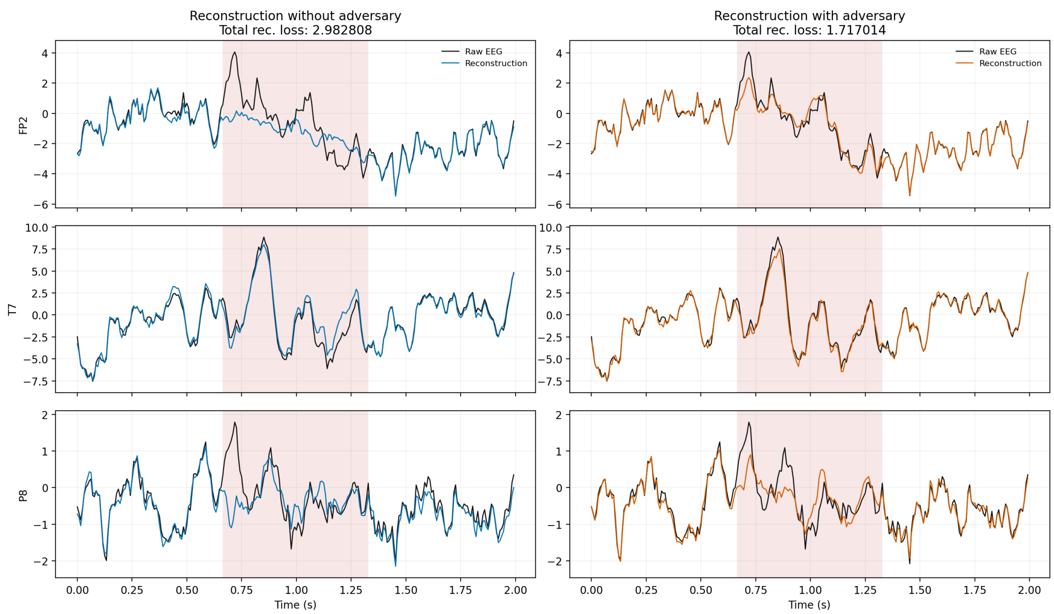

Figure 5: Effect of the adversarial objective on denoising reconstruction. The shaded region marks the masked interval to reconstruct. Adding the discriminator improves the displayed reconstruction loss and preserves sharper EEG morphology, especially around high-amplitude transients.

peaks at the correct frequencies and amplitudes. Without the adversary, the reconstruction can produce plausible-looking oscillations whose peaks are shifted by a few hertz; in this example, the mismatch is especially visible in the 25–35 Hz band. This spectral alignment is important for EEG foundation models, because realistic reconstructed samples should preserve neural signatures as frequency-localized rhythms rather than only minimize pointwise error.

Additional diagnostics show that the improvement is spatially broad and does not destabilize optimization: Appendix Figure 7 reports lower channel-wise reconstruction MSE across the scalp, while Appendix Figure 8 shows close train-validation tracking and stable adversarial losses.

The ablation in Table 3 confirms that the adversarial term is most effective as a regularizer on top of the full denoising objective. Removing the hidden reconstruction term causes a large drop, indicating that adversarial realism on masked patches cannot replace direct prediction of missing content. Using hidden reconstruction with the adversary recovers most of the reconstruction-only performance,

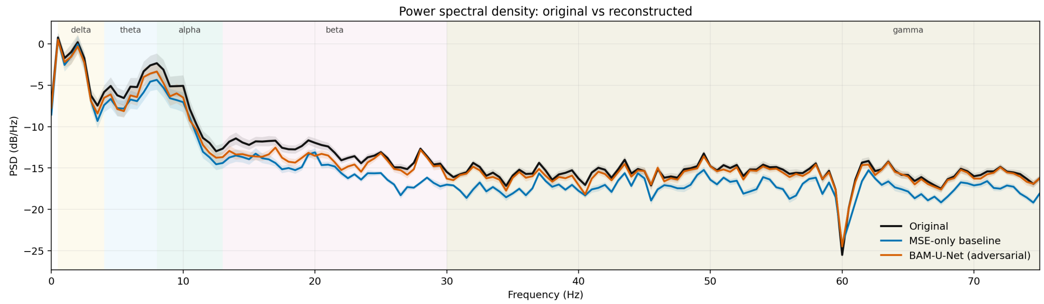

Figure 6: Power spectral density of a reconstructed EEG sample with and without adversarial denoising. The adversarial objective better preserves the frequency and amplitude of spectral peaks, while the non-adversarial reconstruction can shift peaks by a few hertz, notably in the 25–35 Hz range.

Table 3: Ablation of pre-training losses on PhysioNet-MI. Scores are reported in $[0, 1]$; best results are in bold and second-best results are underlined.

Loss combination	**PhysioNet-MI 4-class**		
	Balanced Accuracy	**Cohen's κ**	**Weighted F1**
$\mathcal{L}_{\text{vis}} + \mathcal{L}_{\text{hid}}$	0.5954 ± 0.0290	0.4539 ± 0.0387	0.5958 ± 0.0283
$\mathcal{L}_{\text{vis}} + 0.1\mathcal{L}_{\text{adv}}$	0.4726 ± 0.0418	0.2968 ± 0.0557	0.4718 ± 0.0405
$\mathcal{L}_{\text{hid}} + 0.1\mathcal{L}_{\text{adv}}$	0.5903 ± 0.0310	0.4537 ± 0.0413	0.5909 ± 0.0302
$\mathcal{L}_{\text{vis}} + \mathcal{L}_{\text{hid}} + \mathcal{L}_{\text{adv}}$	$\underline{0.6049 \pm 0.0331}$	$\mathbf{0.4892 \pm 0.0441}$	$\underline{0.6084 \pm 0.0320}$
$\mathcal{L}_{\text{vis}} + \mathcal{L}_{\text{hid}} + 0.1\mathcal{L}_{\text{adv}}$	$\mathbf{0.6141 \pm 0.0337}$	$\underline{0.4853 \pm 0.0448}$	$\mathbf{0.6144 \pm 0.0326}$

while combining visible, hidden, and adversarial losses gives the best downstream accuracy and weighted F1. The selected weight $\lambda_{\text{adv}} = 0.1$ improves balanced accuracy by 1.87 percentage points over reconstruction alone and avoids the overemphasis on adversarial matching suggested by the unit-weighted variant.

5 Limitations and future work

Despite the significant performance gain in linear probing, BAM-U-Net still exhibits limitations. First, despite its effective 1×1 convolutional adapter to enable cross-montage capability, this adapter still requires a small amount of training time to adapt to any montage. Second, Transformer-based architectures exhibit an advantage for anomaly detection as shown with EEGPT being state-of-the-art on the TUAB dataset in linear probing. We aim to extend our framework to be natively montage-agnostic and favor inductive biases because of the still non-existent large-scale high-quality and diverse EEG datasets preventing massive scaling as in NLP or Computer Vision.

6 Conclusion

In this paper, we introduced BAM-U-Net, a bidirectional state-space model that achieves outstanding linear probing performance across diverse EEG downstream tasks, ranking first on five out of six benchmarks and second on the remaining one. Its superior ability to extract robust features without fine-tuning, combined with a flexible framework adaptable to any montage or recording duration, significantly simplifies deployment in real-world scenarios. Furthermore, the linear complexity of Mamba layers, coupled with a lightweight adapter, facilitates adoption by neuroscientists and the broader medical community by drastically reducing computational barriers. We believe our work paves the way for accessible, efficient, and scalable EEG Foundation Models.

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A Foundation models used in the benchmark

We evaluate seven EEG foundation encoders in the linear-probing benchmark. Together, these models span several recent pre-training paradigms, including Transformer-based contextual modeling, masked autoencoding, query-based representation learning, selective state-space sequence modeling, and reconstruction-driven latent learning.

CBraMod. Transformer-based EEG foundation model that encodes channel-wise temporal patches with contextual brain modeling [7].

CSBrain. Transformer-based EEG encoder designed to learn cross-channel and temporal structure from patch-level brain signals [8].

EEGPT. Masked-autoencoder EEG foundation model trained to learn transferable representations from multi-channel EEG windows [5]. We used the LARGE checkpoint.

LaBraM. Large brain model based on patch-tokenized EEG signals and Transformer encoding for general EEG representation learning [6].

LUNA. Query-based EEG foundation encoder trained with masked reconstruction to produce compact transferable signal representations [9].

LuMamba. Mamba-based EEG foundation model combining selective state-space sequence modeling with LeJEPA–reconstruction pre-training [18, 27].

REVE. Reconstruction-based EEG foundation model that learns high-dimensional latent representations from masked EEG windows [4].

B Pre-training ablation

We isolate the contribution of large-scale TUEG pre-training by comparing BAM-U-Net under the same BCIC-2A downstream evaluation protocol with and without the pre-training stage. This ablation uses the 4-class motor imagery setting from BCIC-2A and reports balanced accuracy, Cohen's κ, and weighted F1; the results are reported in Table 4.

Table 4: Effect of TUEG pre-training on BCIC-2A linear probing. Scores are reported in $[0, 1]$.

Setting	**BCIC-2A 4-class**		
	Balanced Accuracy	**Cohen's κ**	**Weighted F1**
Without pre-training	0.5028 ± 0.0324	0.3938 ± 0.0316	0.4916 ± 0.0195
With pre-training	$\mathbf{0.6058 \pm 0.0280}$	$\mathbf{0.4744 \pm 0.0273}$	$\mathbf{0.5923 \pm 0.0168}$

C Additional representation and denoising diagnostics

This section provides secondary diagnostics referenced in Section 4.

Table 5: Summary statistics for pairwise cosine similarity between flattened TUAB embeddings. Lower mean and p_{95} values indicate weaker directional collapse.

Model	Mean	Std.	Cosine p_{95}
CBraMod	0.943	0.102	0.993
CSBrain	0.805	0.142	0.907
EEGPT	0.635	0.049	0.698
LaBraM	0.946	0.027	0.982
LUNA	0.807	0.065	0.893
LuMamba	0.579	0.072	0.697
REVE	0.904	0.097	0.944
BAM-U-Net	**0.517**	0.114	**0.653**

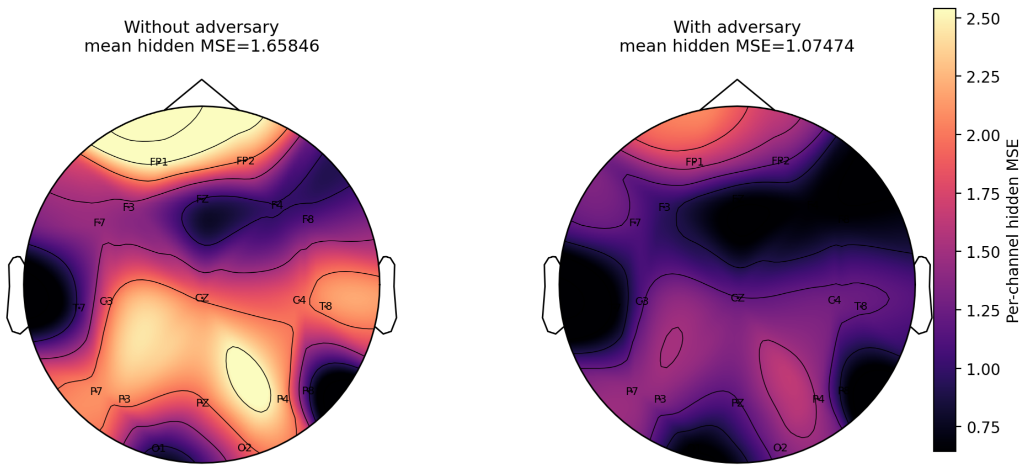

Figure 7: Channel-wise reconstruction MSE topomap with and without adversarial denoising. The adversarial objective improves reconstruction broadly across channels, showing a spatially consistent reduction in error rather than a localized effect.

D Downstream datasets and preprocessing

Evaluation Strategy We chose to evaluate our model on 6 different downstream datasets which span 5 different BCI tasks to ensure the universality of the features extracted and a fair comparison with recent EEG foundation models. We briefly describe the datasets and their uniqueness and detail our preprocessing strategy.

D.1 BCIC-2A

Description The BCIC-2A dataset [28] is a motor imagery EEG dataset, it contains EEG recordings from 9 subjects performing trials of 4 different motor imagery tasks. It was acquired with 22 electrodes placed according to the 10-20 montage system. Two sessions were held on different days, each containing 288 trials.

Evaluation Protocol We employ a strict leave-one-subject-out (LOSO) cross-validation strategy. Specifically, the frozen encoder extracts features for all recordings, and a linear classifier is trained on eight subjects before being tested on the held-out ninth subject. This process is iterated nine times to cover all participants, and we report the averaged test performance in Table 6.

D.2 PhysioNet-MI

Description The PhysioNet-MI dataset [29, 30] is a motor imagery EEG dataset, it contains EEG recordings from 109 subjects performing 4 different motor imagery tasks. It was acquired with 64 electrodes placed according to the 10-10 montage.

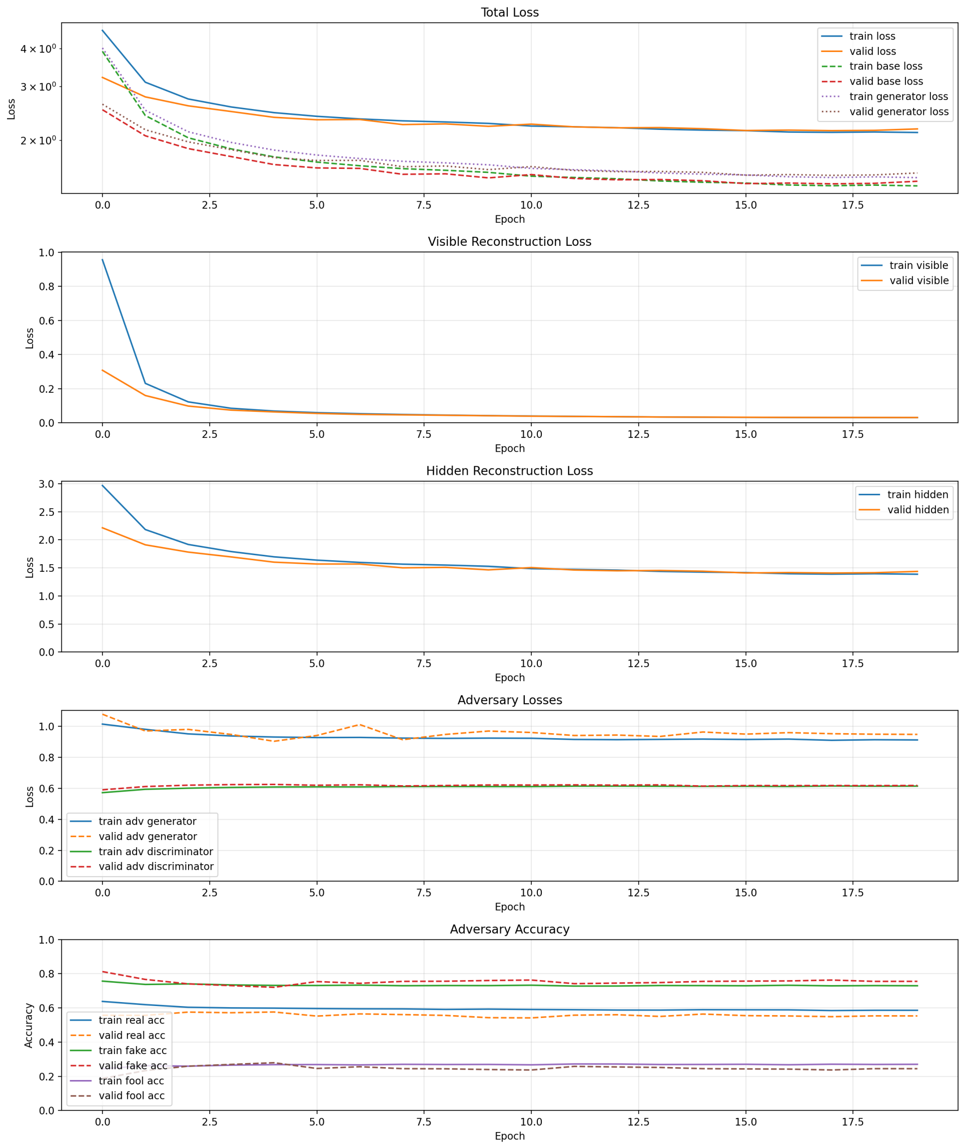

Figure 8: Pre-training dynamics of BAM-U-Net. Reconstruction losses decrease smoothly with close train-validation tracking, while the adversarial generator and discriminator terms remain stable rather than saturating.

Table 6: The results of different methods on motor imagery classification (BCIC-2A, 4-class).

Methods	Balanced Accuracy	Cohen's Kappa	Weighted F1
CBraMod	0.2753 ± 0.0196	0.0338 ± 0.0262	0.1630 ± 0.0474
CSBrain	0.4719 ± 0.0005	0.2958 ± 0.0007	0.4555 ± 0.0006
EEGPT	$\underline{0.5327 \pm 0.0151}$	$\underline{0.3770 \pm 0.0202}$	$\underline{0.5141 \pm 0.0164}$
LaBraM	0.3130 ± 0.0004	0.0840 ± 0.0005	0.2810 ± 0.0008
LUNA	0.3373 ± 0.0390	0.1164 ± 0.0520	0.3155 ± 0.0484
LuMamba	0.3169 ± 0.0378	0.0892 ± 0.0503	0.2984 ± 0.0419
REVE	0.5181 ± 0.0939	0.3575 ± 0.1252	0.4985 ± 0.1139
BAM-U-Net	$\mathbf{0.6058 \pm 0.0280}$	$\mathbf{0.4744 \pm 0.0273}$	$\mathbf{0.5923 \pm 0.0168}$

Evaluation Protocol We utilize the dataset's official test split for our final evaluation. To optimize the linear probe without data leakage, we partition the training cohort into 5 distinct folds. A 5-fold subject-wise cross-validation is then performed to train the classification head on the frozen representations, with results shown in Table 7.

Table 7: The results of different methods on motor imagery classification (PhysioNet-MI, 4-class).

Methods	Balanced Accuracy	Cohen's Kappa	Weighted F1
CBraMod	0.3978 ± 0.0562	0.1971 ± 0.0751	0.3570 ± 0.0870
CSBrain	0.5114 ± 0.0325	0.3485 ± 0.0433	0.5097 ± 0.0314
EEGPT	$\underline{0.5467 \pm 0.0221}$	$\underline{0.3956 \pm 0.0295}$	$\underline{0.5487 \pm 0.0208}$
LaBraM	0.2782 ± 0.0055	0.0376 ± 0.0073	0.2683 ± 0.0084
LUNA	0.4314 ± 0.0216	0.2418 ± 0.0288	0.4289 ± 0.0212
LuMamba	0.4411 ± 0.0217	0.2548 ± 0.0289	0.4412 ± 0.0216
REVE	0.4719 ± 0.0450	0.2958 ± 0.0933	0.4714 ± 0.0461
BAM-U-Net	$\mathbf{0.6183 \pm 0.0317}$	$\mathbf{0.4909 \pm 0.0423}$	$\mathbf{0.6190 \pm 0.0316}$

D.3 KaggleERN

Description The KaggleERN dataset [31] is an error-related potential dataset containing EEG recordings of subjects exposed to flashing items. Each subject fixes an item, then the system displays the chosen item or another one. Therefore, subjects observe whether the system's response to a P300-speller task is correct or erroneous. The objective is to decode Error-Related Potentials (ErrPs) to classify the feedback as correct or incorrect.

Evaluation Protocol Following standard benchmark practices, the official independent test set is reserved exclusively for final evaluation. The training phase of our linear adapter is conducted using a 5-fold subject-wise cross-validation approach on the remaining data, ensuring that features from the same subject do not overlap between the training and validation splits. Table 8 reports the corresponding results.

Table 8: The results of different methods on error-related potential classification (KaggleERN, 2-class).

Methods	Balanced Accuracy	AUPRC	AUROC
CBraMod	0.5002 ± 0.0014	0.7223 ± 0.0087	0.5081 ± 0.0199
CSBrain	0.5428 ± 0.0014	0.7692 ± 0.0009	0.5814 ± 0.0019
EEGPT	$\underline{0.5575 \pm 0.0074}$	$\mathbf{0.7847 \pm 0.0044}$	$\mathbf{0.6194 \pm 0.0042}$
LaBraM	0.4998 ± 0.0001	0.7019 ± 0.0075	0.4830 ± 0.0124
LUNA	0.5108 ± 0.0019	0.7224 ± 0.0021	0.5205 ± 0.0038
LuMamba	0.5426 ± 0.0089	0.7527 ± 0.0067	0.5744 ± 0.0111
REVE	0.5520 ± 0.0041	0.7601 ± 0.0092	0.5846 ± 0.0190
BAM-U-Net	$\mathbf{0.5641 \pm 0.0034}$	$\underline{0.7769 \pm 0.0072}$	$\underline{0.6152 \pm 0.0122}$

D.4 Sleep-EDFx

Description The Sleep-EDFx dataset [32] is a sleep staging classification dataset. It contains EEG recordings from 78 healthy subjects while sleeping. It consists of 5 classes corresponding to the sleep stages W, N1, N2, N3, and REM.

Evaluation Protocol Since Sleep-EDFx does not provide a predefined test set, we adopt a 5-fold subject-wise cross-validation protocol. The 78 subjects are divided into 5 mutually exclusive folds. In each iteration, the linear head is evaluated on one held-out fold while being trained and validated on the remaining data (using a standard 9:1 train-validation split). Table 9 reports the corresponding results.

Table 9: The results of different methods on sleep staging classification (Sleep-EDFx, 5-class). Best results are in bold and second-best results are underlined.

Methods	Balanced Accuracy	Cohen's Kappa	Weighted F1
CBraMod	0.5091 ± 0.1162	0.4692 ± 0.1378	0.5741 ± 0.1126
CSBrain	0.5817 ± 0.0143	0.5430 ± 0.0114	0.6456 ± 0.0130
EEGPT	0.6010 ± 0.0172	0.5930 ± 0.0310	0.6794 ± 0.0225
LaBraM	0.6252 ± 0.0470	0.6023 ± 0.0575	0.6988 ± 0.0507
LUNA	0.6002 ± 0.0420	0.5798 ± 0.0550	0.6708 ± 0.0458
LuMamba	$\underline{0.6565 \pm 0.0345}$	$\underline{0.6563 \pm 0.0457}$	$\underline{0.7380 \pm 0.0357}$
REVE	0.6087 ± 0.0472	0.5667 ± 0.0609	0.6813 ± 0.0449
BAM-U-Net	$\mathbf{0.6774 \pm 0.0186}$	$\mathbf{0.6727 \pm 0.0205}$	$\mathbf{0.7469 \pm 0.0249}$

D.5 TUAB

Description The TUAB dataset [33] is an abnormality detection dataset. It contains 409,455 ten-second EEG recordings annotated as either normal or abnormal.

Evaluation Protocol Final metrics are computed directly on the official TUAB evaluation set. For the linear probing phase, the training participants are split into 4 folds. A 4-fold subject-wise cross-validation is applied to train the linear classification head on top of our frozen BAM-U-Net encoder, maintaining a strict separation of subjects. Table 10 reports the corresponding results.

Table 10: The results of different methods on abnormal EEG detection (TUAB, 2-class).

Methods	Balanced Accuracy	AUPRC	AUROC
CBraMod	0.6455 ± 0.0765	0.6551 ± 0.1225	0.6942 ± 0.1006
CSBrain	0.7601 ± 0.0023	0.8462 ± 0.0012	0.8409 ± 0.0011
EEGPT	$\mathbf{0.7981 \pm 0.0190}$	$\mathbf{0.8824 \pm 0.0200}$	$\mathbf{0.8772 \pm 0.0183}$
LaBraM	0.7748 ± 0.0007	0.8603 ± 0.0006	0.8579 ± 0.0007
LUNA	0.6989 ± 0.0017	0.7666 ± 0.0019	0.7681 ± 0.0019
LuMamba	0.7356 ± 0.0053	0.8138 ± 0.0045	0.8249 ± 0.0011
REVE	0.7274 ± 0.0175	0.8010 ± 0.0125	0.8188 ± 0.0084
BAM-U-Net	$\underline{0.7928 \pm 0.0081}$	$\underline{0.8690 \pm 0.0160}$	$\underline{0.8707 \pm 0.0075}$

D.6 TUEV

Description The TUEV dataset [34, 33] is a seizure detection dataset derived from TUEG. It contains 6 classes: spike and sharp wave (SPSW), generalized periodic epileptiform discharges (GPED), periodic lateralized epileptiform discharges (PLED), eye movement (EYEM), artifact (ARTF), and background (BCKG).

Evaluation Protocol Similar to TUAB, we report our final results on the predefined TUEV evaluation split. The linear adapter is optimized using a 4-fold subject-wise cross-validation strategy exclusively on the training set, guaranteeing robust feature transferability assessment across different patients. Table 11 reports the corresponding results.

Preprocessing strategy To ensure transferability between pre-training and downstream tasks, all the downstream datasets are preprocessed and resampled to 200 Hz. Preprocessing is adapted to each dataset corresponding to their unique challenges. We strictly follow the same preprocessing as in EEGPT [5] to remove any potential bias and ensure that our benchmarks remain directly comparable with the current state-of-the-art. By maintaining this consistency, we guarantee that the observed performance improvements are strictly attributable to our model's architecture and pre-training strategy rather than variations in signal quality or data cleaning. Detailed parameters for each dataset, including specific frequency cut-offs and windowing, are provided in Table 12.

Table 13 aggregates the average performance across all evaluated downstream datasets and metrics.

Table 11: The results of different methods on seizure/event detection (TUEV, 6-class).

Methods	**Balanced Accuracy**	**Cohen's Kappa**	**Weighted F1**
CBraMod	0.4422 ± 0.0122	0.4985 ± 0.0189	0.7358 ± 0.0069
CSBrain	$\underline{0.5236 \pm 0.0100}$	$\mathbf{0.5684 \pm 0.0202}$	$\underline{0.7727 \pm 0.0098}$
EEGPT	0.4213 ± 0.0122	0.4197 ± 0.0062	0.6805 ± 0.0062
LaBraM	0.4409 ± 0.0107	0.5224 ± 0.0113	0.7476 ± 0.0071
LUNA	0.2719 ± 0.0110	0.2680 ± 0.0265	0.6647 ± 0.0096
LuMamba	0.3222 ± 0.0332	0.3976 ± 0.0445	0.7152 ± 0.0140
REVE	0.4414 ± 0.0323	0.4720 ± 0.0610	0.7463 ± 0.0365
BAM-U-Net	$\mathbf{0.5258 \pm 0.0193}$	$\underline{0.5655 \pm 0.0156}$	$\mathbf{0.7916 \pm 0.0056}$

Table 12: Detailed preprocessing and windowing parameters per dataset.

Dataset	**Filter Range**	**Window**	**Samples**	**Evaluation Protocol**
BCIC-2A	0.3 – 38 Hz	4s	800	LOSO (9-fold)
PhysioNet-MI	> 0.3 Hz	4s	800	5-fold Subject-wise
KaggleERN	–	2s	400	5-fold Subject-wise
Sleep-EDFx	< 30 Hz	30s	6000	5-fold Subject-wise
TUEV	0.1 – 75 Hz*	5s	1000	4-fold Subject-wise
TUAB	0.1 – 75 Hz*	10s	2000	4-fold Subject-wise

Table 13: Average performance across the evaluated datasets. Balanced Accuracy is averaged across all 6 datasets. Cohen's Kappa and Weighted F1 are averaged across 4 datasets, while AUPRC and AUROC are averaged across 2 datasets. Standard deviations are omitted for clarity. Best results are in bold.

Method	**Balanced Acc.**	**Cohen's Kappa**	**Weighted F1**	**AUPRC**	**AUROC**
CBraMod	0.4617	0.2996	0.4575	0.6887	0.6012
CSBrain	0.5652	0.4389	0.5959	0.8077	0.7111
EEGPT	<u>0.5762</u>	<u>0.4463</u>	<u>0.6057</u>	**0.8336**	**0.7483**
LaBraM	0.4886	0.3116	0.4989	0.7811	0.6705
LUNA	0.4751	0.3015	0.5200	0.7445	0.6443
LuMamba	0.5025	0.3495	0.5482	0.7833	0.6997
REVE	0.5533	0.4230	0.5994	0.7806	0.7017
BAM-U-Net	**0.6307**	**0.5509**	**0.6875**	<u>0.8230</u>	<u>0.7430</u>

E Computational scaling with sequence length

We measure the number of floating-point operations required by each encoder as a function of the input sequence length. Figure 9 reports the FLOPs per inference in float32. BAM-U-Net shows an approximately linear increase over the evaluated range. LuMamba follows a similar trend and uses fewer FLOPs than BAM-U-Net at the measured sequence lengths. By contrast, the Transformer-based baselines generally start from higher FLOP counts and grow more rapidly, with REVE and EEGPT being the most computationally expensive models in this comparison. These measurements support the computational advantage of Mamba-based architectures for long EEG sequences.

F Implementation Details

In this section, we report the pre-training hyperparameters and the computational resources used in the experimental settings of the BAM-U-Net model. Table 14 summarizes the architecture, optimization, and compute settings.

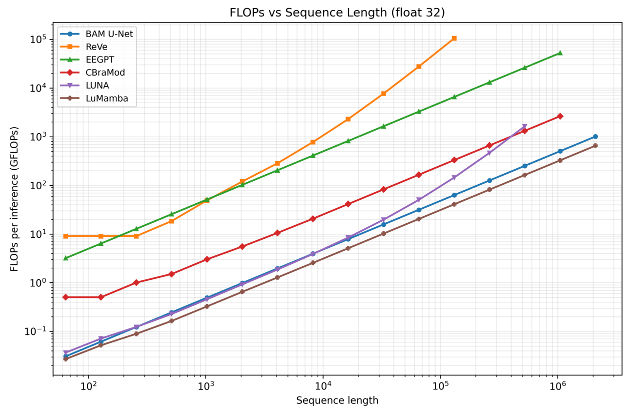

Figure 9: FLOPs as a function of sequence length. FLOPs per inference are reported in float32 for all foundation models.

Table 14: Detailed architecture and training hyperparameters of BAM-U-Net.

Component	**Parameter**	**Value**
Temporal Embedder	Kernel sizes	[1, 3, 7, 11, 51, 101, 201]
	Virtual Channels	32
Encoder	Number of stages	4
	Blocks per stage	3
	Downsampling	MaxPool1D
Decoder	Number of stages	4
	Blocks per stage	3
	Upsampling	Linear interpolation
Scheduler	Type	WarmupCosineLR
	Max Learning Rate (η_{max})	5×10^{-4}
	Warm-up ratio	0.1
	Min Learning Rate (η_{min})	0.0
	Weight Decay	0.01
Optimizer	Type	AdamW
	Epsilon	1e-8
	Betas	[0.9, 0.999]
Pre-training learning dynamics	Training Epochs	20
	Batch size	32
Hardware & Compute	GPU Type	NVIDIA A100 (40GB)
	Number of GPUs	1
	Total Pre-training Time	27 hours
Software Stack	Programming Language	Python 3.12.12
	Deep Learning Framework	PyTorch 2.7.1
	CUDA Version	12.9
	`mamba-ssm`	2.2.6

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The claims in the abstract and introduction accurately reflect the paper's contributions. The introduction highlights an adversarial denoising objective, state-of-the-art performance in linear probing on 5/6 datasets, and linear-time sequence modeling. The following sections provide empirical evidence for these claims:

   - We introduce an adversarial network (Section 2.5) which enhances feature quality, as shown in our ablations.
   - We compare BAM-U-Net against 7 baselines and show that our model outperforms them on 5 out of 6 datasets and ranks second on the remaining one (Section 4).
   - We demonstrate that the BiMamba-2 backbone achieves linear-time complexity (Section E of the Appendix) with respect to the sequence length.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
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   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
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2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: In Section 5 we clearly mention that BAM-U-Net is constrained by its small cross montage adapter and that Mamba-based models tend to underperform on anomaly detection.

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